POWER BI

What is BI?

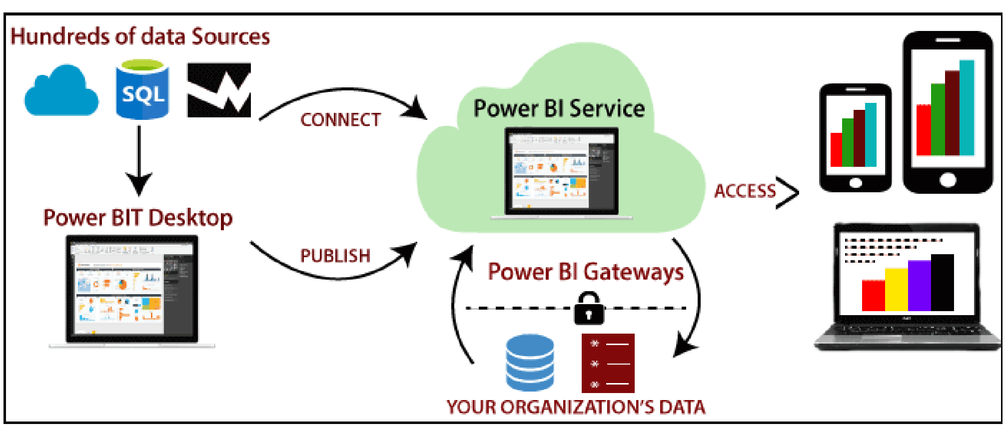
The BI term refers to **Business Intelligence**. It is a data-driven decision support system (DSS), which helps you to analyze the data and provide actionable information. It helps the business manager, corporate executives, and other users in making their decisions easily.

Business intelligence refers to the applications, technologies, and practices for the collection, analysis, integration, and presents the business information. The purpose of business intelligence is to support better decision making.

What is Power BI?

Power BI is a **Data Visualization**, and **Business Intelligence** tool which helps to convert data from different data sources into interactive dashboards and BI reports. It provides interactive visualizations with self-service business intelligence capabilities where end users can create reports and dashboards by themselves, without having to depend on information technology staff or database administrators.

Power BI provides multiple connectors, software, and services. These services based on the **SaaS** and mobile Power BI apps which are available for different platforms. These set of services are used by business users to consume data and to build BI



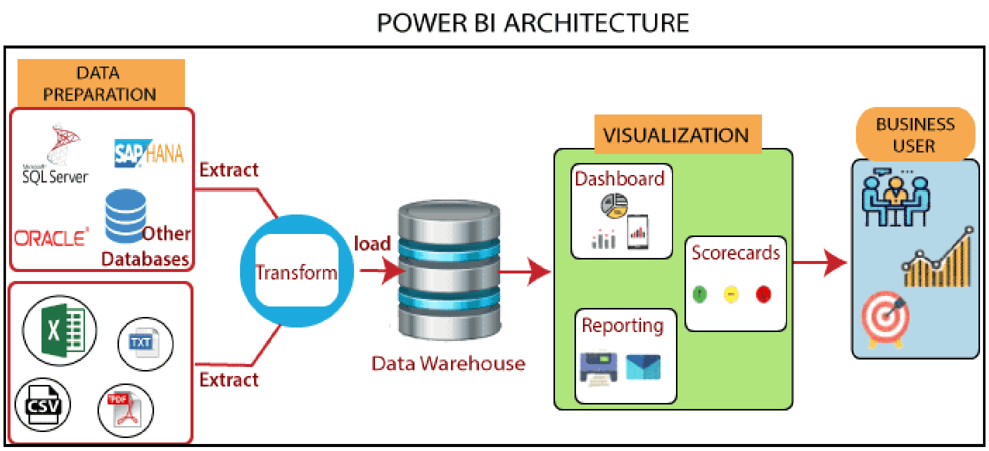
reports.

Power BI desktop app is used to create reports, while Power BI Service (Software as a Service - SaaS) is used to publish those reports. And Power BI mobile app is used to view the reports and dashboards.

Different Power BI version like Desktop, Service-based (SaaS), and mobile Power BI apps are used in different platforms.

Power BI Architecture

The architecture of Power BI is shown as below:



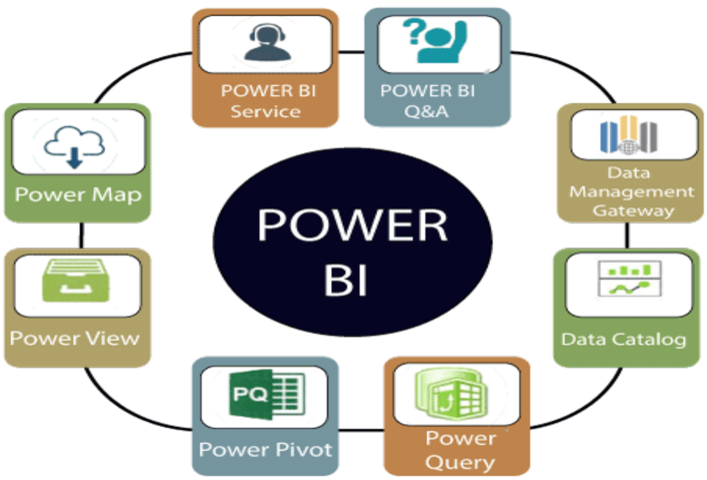
Power BI architecture has three phases. The first two phases use ETL (extract, transform, and load) process to handle the data.

1. **Data Integration:** An organization needs to deal with the data that comes from differentsources.First, extract the data from different sources which can be yourseparatedatabase,servers,etc.Then the data is integrated into a standard format and stored at a common area that's called staging area.
2. **Data Processing:** Still, the integrated data is not ready for visualization because the data needs processing before it can be presented. This data is pre-processed.**For example,** the missing values or redundant values will be removed from the datasets.
3. After that, the business rules will be applied to the data, and it transforms into presentabledata.Then this data will be loaded into the data warehouse.
4. **Data presentation:** Once the data is loaded and processed, then it can be visualized much better with the use of various visualization that Power BI offers.By using of dashboard and reports, we represent the data more

intuitively.These visual reports help business end-users to take business decision based on the insights.

Power BI Components

The components of Power BI are shown as below:



1. Power Query: It is used to access, search, and transform public and internal data sources.

2. Power Pivot: Power pivot is used in data modeling for in-memory analytics.

3. Power View: By using the power view, you can analyze, visualize, and display the data as an interactive data visualization.

4. Power Map: It brings the data to life with interactive geographical visualization.

5. Power BI Service: You can share workbooks and data views which are restored from on-premises and cloud-based data sources.

6. Power BI Q&A: You can ask any questions and get an immediate response with the natural language query.

7. Data Management Gateway: You get periodic data refreshers, expose tables, and view data feeds.

8. Data Catalog: By using the data catalog, you can quickly discover and reuse the queries.

Power BI Advantages

1. Secure Report Publishing: You can automate setup data refresh and publish reports that allowing all the users to avail the latest information.

2. No Memory and Speed Constraints: To Shift an existing BI system into a powerful cloud environment with Power BI embedded eliminates memory. Speed constraints ensure that data is quickly retrievable and analyzed.

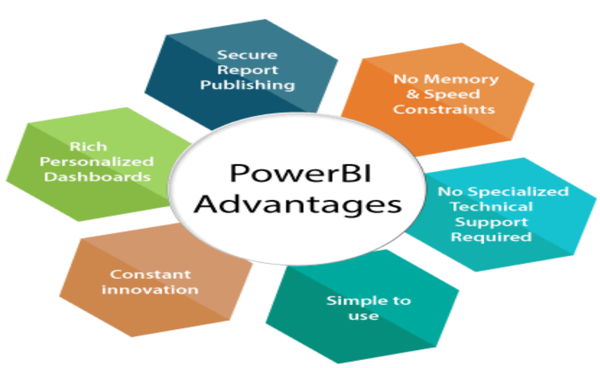
3. No Specialized Technical Support required: The Power BI provides quick inquiry and analysis without the need for specialized technical support. It also supports a powerful natural language interface and the use of intuitive graphical designer tools.

4. Simple to Use: Power BI is simple to use. Users can easily find it only on behalf of a short learning curve.

5. Constant innovation: The Power BI product is updated in every month with new functions and features.

6. Rich, personalized dashboard: The crowning feature of Power BI is the information dashboards that can be customized to meet the exact need of any enterprise. You can easily embed the dashboards, and BI reports in the applications to provide a unified user experience.

Here are some advantages of Power BI, as shown below:



Power BI Disadvantages

Here are some disadvantages of Power BI, as shown below:

1. Dashboards and reports are only shared with the users who are having the same email domains.
2. Power BI will not merge imported data that is accessed from real-time connections.
3. Power BI only accepts the file size maximum 250 Mb and the zip file which is compressed by the data of the x-velocity in-memory database.
4. Dashboard never accepts or pass user, account, or any other entity parameters.
5. Very few data sources permit real-time connections to Power BI reports and dashboards.

Download and Install Power BI Desktop

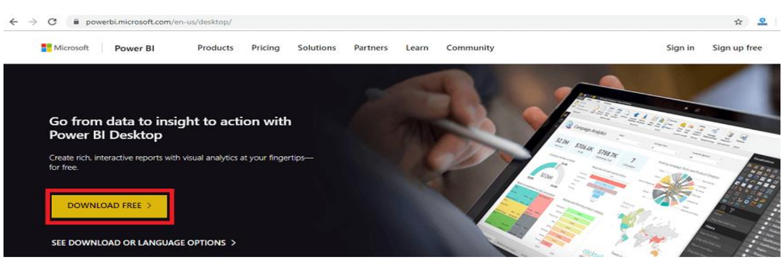
Here are some requirements of the system to download the Power BI Desktop:

1. Window 7, window 8, window 8.1, window 10, and windows server 2008 R2, windows server 2012, windows server 2012 R2.
2. It requires internet explorer 9 or higher.
3. Power BI Desktop is available for both 32 bit and 64-bit platforms.

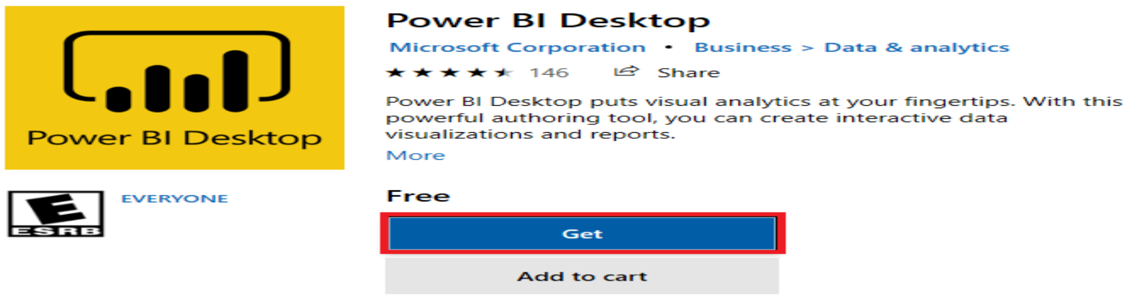
Let's see the downloading process of the Power BI Desktop step by step:

Step 1: Click on the below link to directly download Power BI Desktop. https://powerbi.microsoft.com/en-us/desktop/

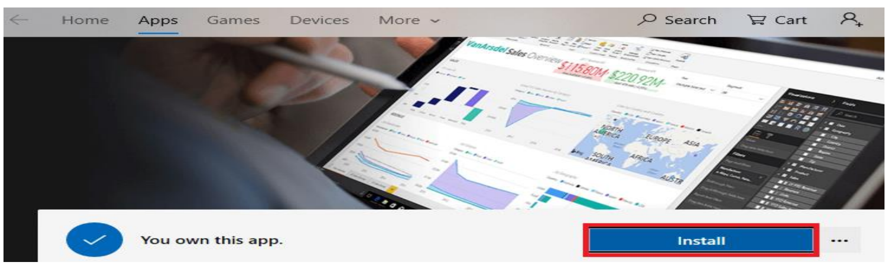
Step 2: Then click on the Download Free button.



Step 3: Now, you will redirect to a Microsoft Store and then select the Get button.

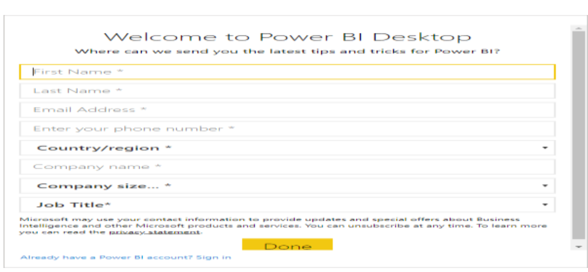


Step 4: Click on the Install button.

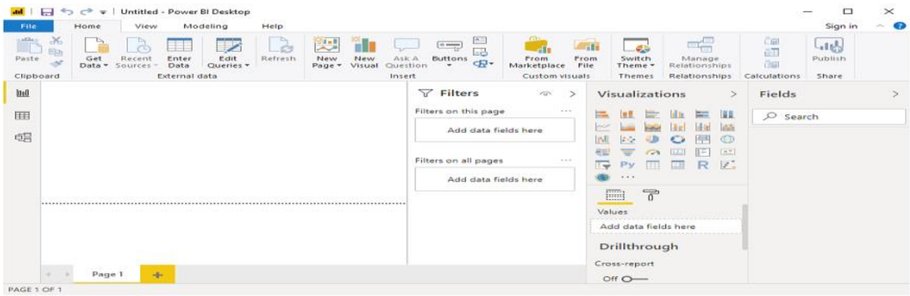


You can see the progress status of the Power BI Desktop on the screen.

Step 5: You can see "welcome to Power BI Desktop" screen and then register yourself on the desktop.



Step 6: When you run the Power BI desktop, it displayed the home page or welcome screen.

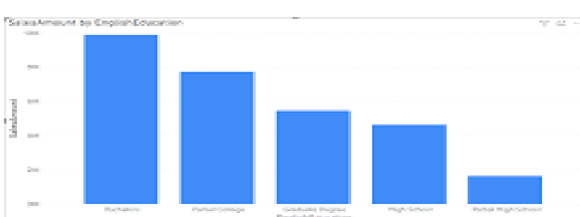


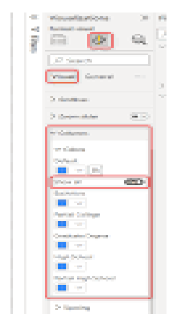
CHARTS

In Business Intelligence (BI), charts are essential tools for visually representing data insights and trends. They help stakeholders understand complex information quickly and make informed decisions. BI platforms typically offer various types of charts to cater to different data visualization needs.

1. Column Chart:

- ✓ Column charts are used to compare data across categories by displaying vertical bars. Each bar represents a category, and the height of the bar represents the value of the data.

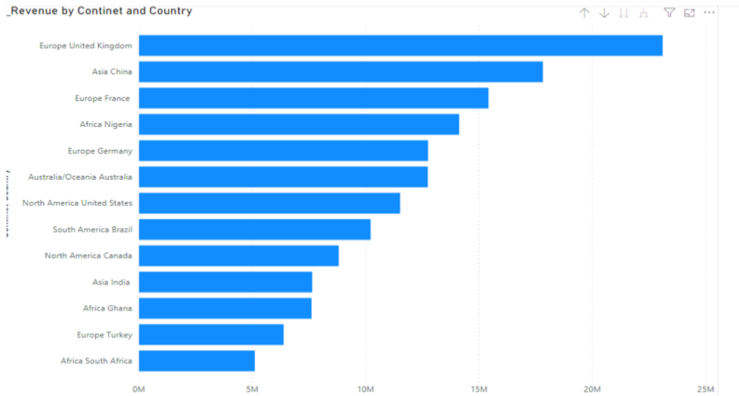




- ✓ **Use**-Comparing sales performance by product, analyzing revenue by region, visualizing monthly expenses, etc.

2. Bar Chart:

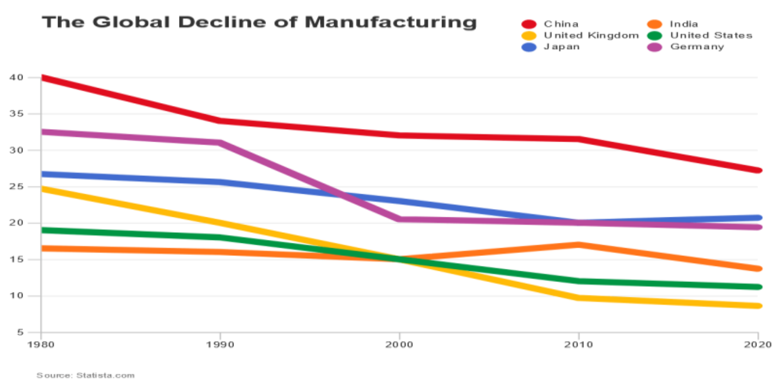
- ✓ Similar to column charts, bar charts compare data across categories, but with horizontal bars instead of vertical bars. The length of the bar represents the value of the data.



- ✓ **Use**-Comparing market share by company, visualizing survey responses by option, analyzing project timelines, etc.

3. Line Chart:

- ✓ Line charts display data points connected by lines, making them suitable for showing trends over time. They are commonly used to visualize time-series data.



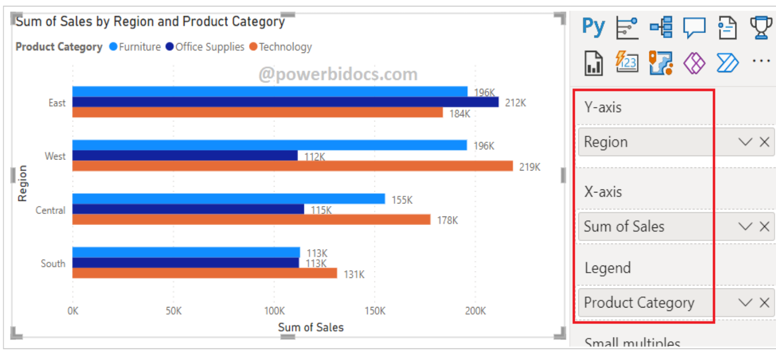
- ✓ **Use**-Analyzing stock prices over time, tracking website traffic trends, monitoring temperature changes, etc.

4. Area Chart:

- ✓ Area charts are similar to line charts but with the area below the lines filled with color. They are used to visualize cumulative totals over time or to show the proportion of different categories over time.
- ✓ **Use-**Visualizing cumulative sales over a period, comparing market share trends, tracking changes in portfolio value, etc.

5. Clustered Bar Chart:

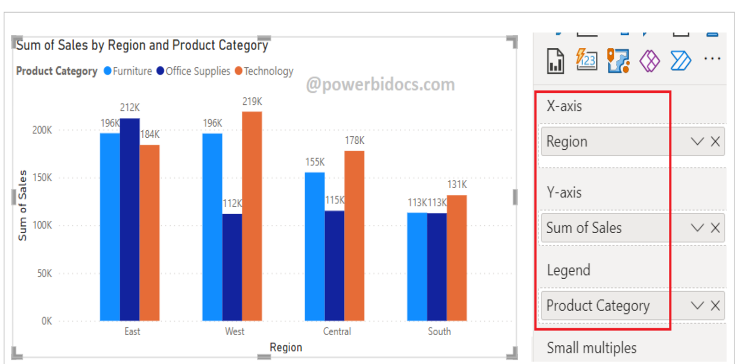
- ✓ Displays data in rectangular bars grouped by category, with each group represented by a cluster of bars.



- ✓ **Use-**For comparing values across different categories.
- ✓ Can be horizontal or vertical.

6. Clustered Column Chart:

- ✓ Similar to a clustered bar chart but uses vertical columns instead of bars.



- ✓ Each cluster of columns represents a group, and each column within the cluster represents a category.

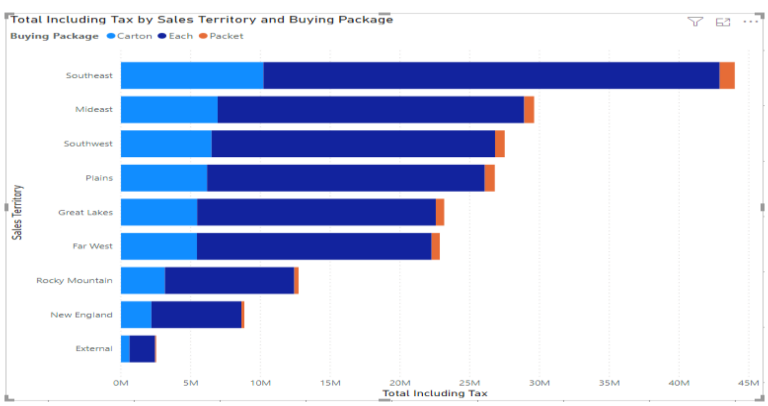
7. 100% Stacked Bar Chart:

✓ Displays data in bars where the lengths of the bars represent the percentage of the whole.

✓ Each bar is divided into segments, and the total length of each bar is normalized to 100%.

8. 100% Stacked Column Chart:

✓ Similar to a 100% stacked bar chart but uses vertical columns instead of bars.



✓ Each column is divided into segments, and the total height of each column is normalized to 100%.

9. Stacked Area Chart:

✓ Represents data using filled areas, with each area stacked on top of the others.

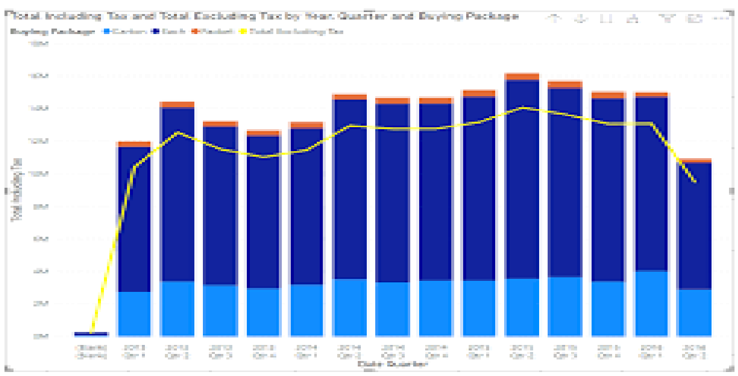
✓ **Use-** For showing how different categories contribute to the total over time.

10. 100% Stacked Area Chart:

✓ Similar to a stacked area chart but with the total area normalized to 100%.
✓ Shows the percentage contribution of each category to the total over time.

11. Line & Stacked Column Chart:

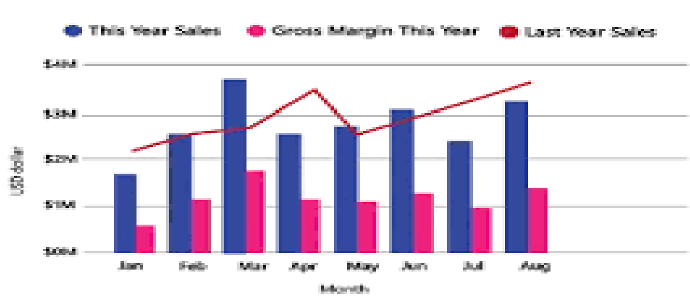
✓ Combines a line chart and a stacked column chart in the same visualization.



✓ **Use-** For comparing trends over time along with the contribution of different categories to the total.

12. Line & Clustered Column Chart:

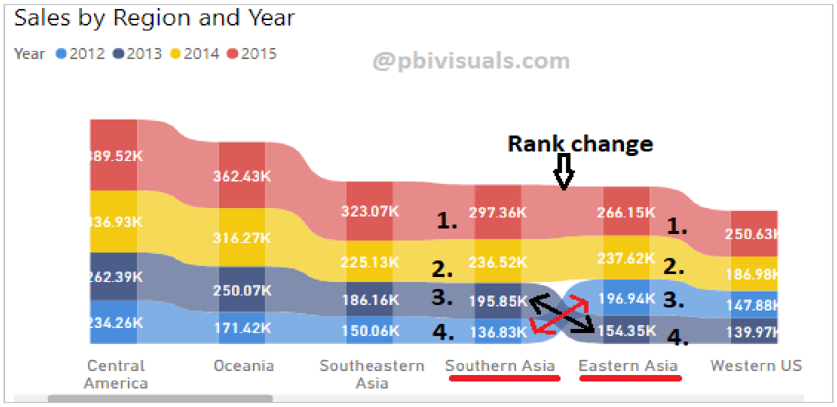
✓ Similar to the line & stacked column chart but uses clustered columns instead of stacked columns.



✓ Each category is represented by a separate cluster of columns, allowing for easier comparison

13. Ribbon Chart:

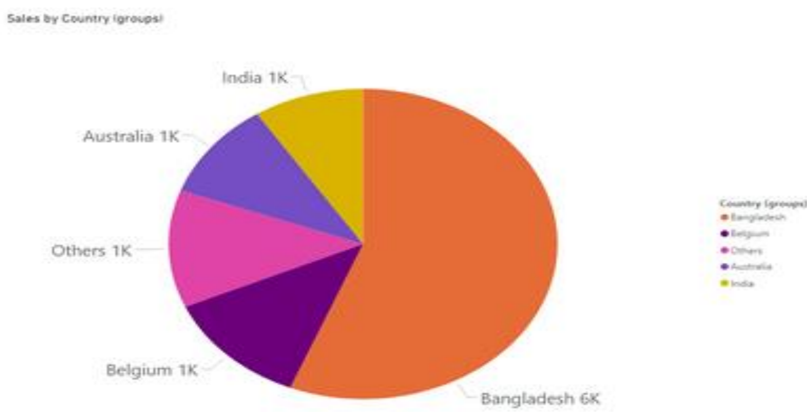
- ✓ Visualizes data over time, with each ribbon representing a category.



- ✓ **Use**-Showing revenue trends for multiple product categories over a period.

14. Pie Chart:

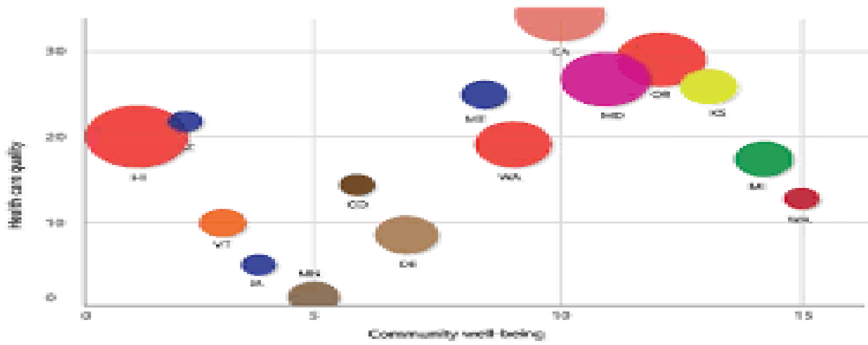
- ✓ Pie charts display data as slices of a circular pie, with each slice representing a proportion of the whole. They are used to show the composition of a dataset.



- ✓ **Use-**Showing the distribution of sales by product category, visualizing market share by company, analyzing expenditure breakdown, etc.

15. Scatter Plot:

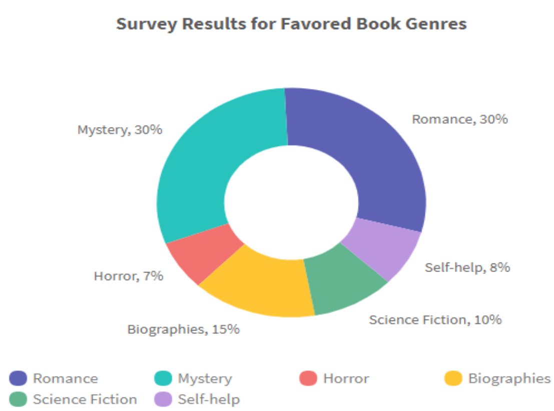
- ✓ Scatter plots display individual data points as dots on a graph, with each dot representing a combination of two variables. They are used to identify relationships between variables.



- ✓ **Use-**Analyzing the relationship between temperature and ice cream sales, studying the correlation between advertising expenditure and sales revenue, etc.

16. Donut Chart:

- ✓ Donut charts are similar to pie charts but with a hole in the center. They are used to show the same kind of information as pie charts but are often preferred for displaying data with multiple categories.



- ✓ **Use**-Visualizing sales distribution by region, analyzing expenditure breakdown by category, etc.

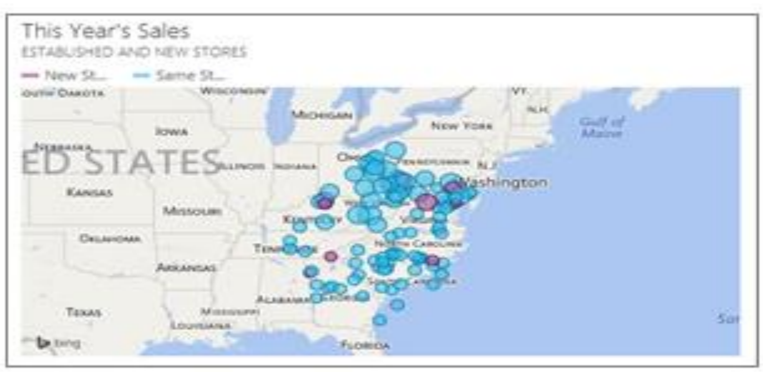
17. Gauge Chart:

- ✓ Gauge charts display a single value within a range of values, represented as a dial or needle on a gauge. They are useful for showing progress towards a goal or target.

- ✓ **Use**-Monitoring project progress, tracking KPIs, visualizing sales targets, etc.

18. Map:

- ✓ The Map visual in Power BI displays data on a map, allowing users to visualize geographic distributions and spatial relationships. It's ideal for showing regional data such as sales by country or customer distribution.



- ✓ **Use**-Visualizing customer locations, analyzing sales performance by region, tracking delivery routes, etc.

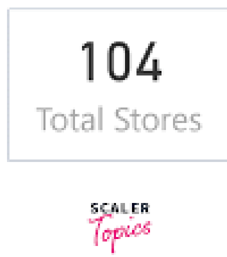
19. Filled Map:

- ✓ Filled Map is similar to the Map visual but fills regions with color based on data values. It's useful for visualizing density or concentration of data across regions.

- ✓ **Use**-Displaying population density, visualizing sales density by state, analyzing crime rates by city, etc.

20. Info Card:

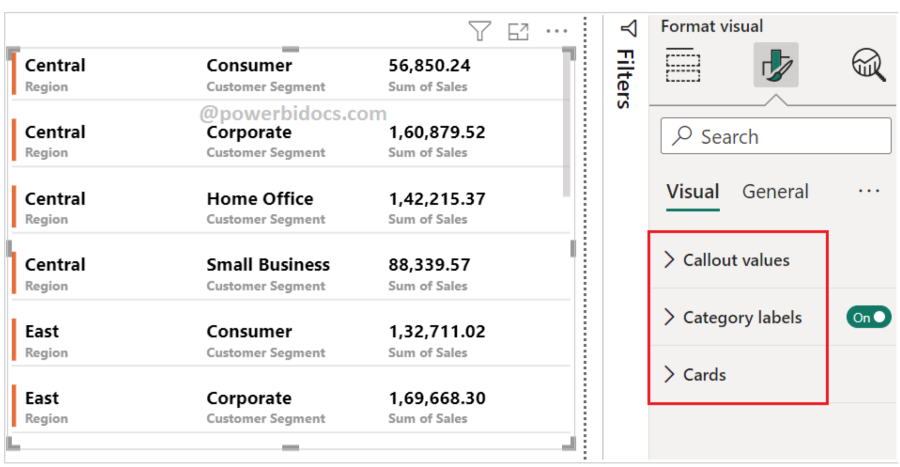
- ✓ Info Card displays a single metric or KPI (Key Performance Indicator) value prominently. It's useful for highlighting key numbers or metrics without additional context.



- ✓ **Use**-Displaying total sales revenue, showing current stock price, visualizing monthly website visitors, etc.

21. Multi Info Card:

- ✓ Multi Info Card is similar to Info Card but allows users to display multiple metrics or KPIs side by side. It's useful for comparing multiple key numbers or metrics at a glance.



- ✓ **Use**-Comparing revenue and expenses, displaying key performance indicators for different departments, visualizing sales targets vs. actuals, etc.

22. KPI (Key Performance Indicator):

- ✓ KPI visual displays a single value along with a target value and a trend indicator (such as up or down arrows) to show progress towards a goal. It's useful for monitoring performance against predefined targets.

- ✓ **Use**-Tracking sales performance against targets, monitoring website traffic metrics, visualizing project completion rates, etc.

23. Slicer:

- ✓ Slicer allows users to filter data in other visuals based on selected criteria. It's useful for interactive filtering and drilling down into specific subsets of data.
- ✓ **Use**-Filtering sales data by date range, filtering product categories, selecting specific regions on a map, etc.

24. Table:

TABELA: Top 10 weather states with afford Y IE

Estado	Acessibilidade	Clima	Classificação geral
Havaí	45	1	10
Flórida	25	2	5
Louisiana	29	3	36
Texas	24	4	17
Geórgia	19	5	28
Mississippi	6	6	19
Alabama	10	7	16
Carolina do Sul	27	8	41
Arkansas	4	9	11
Arizona	33	10	38

- ✓ The Table visual displays data in tabular format, similar to a spreadsheet. It's useful for viewing detailed data and comparing values across different categories or dimensions.

- ✓ **Use**-Displaying raw data, showing detailed sales transactions, visualizing survey responses, etc.

25. Matrix:

- ✓ The Matrix visual in Power BI is similar to a pivot table in Excel. It allows users to display data in a tabular format with rows and columns, and it supports grouping, subtotaling, and aggregating data. Users can drag and drop fields to define rows, columns, and values.

Year	2013		2014	
Segment	Total Sales	Total Profit	Total Sales	Total Profit
Channel Partners	3,98,090.28	2,89,889.28	14,02,503.36	10,26,913.86
Enterprise	40,49,562.50	-1,93,757.50	1,55,62,131.88	-4,20,788.13
Government	1,30,85,685.28	28,86,645.28	3,94,18,575.39	85,01,527.89
Midmarket	5,46,243.45	1,51,763.45	18,35,639.63	5,08,339.63
Small Business	83,35,674.00	7,43,924.00	3,40,92,244.50	33,99,244.50
Total	**2,64,15,255.51**	**38,78,464.51**	**9,23,11,094.75**	**1,30,15,237.75**

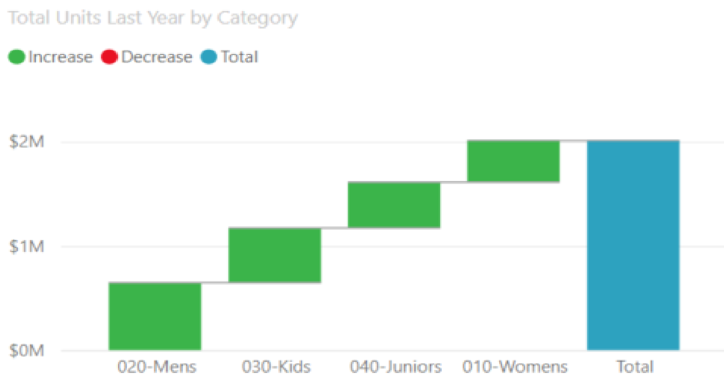
- ✓ **Use**-The Matrix visual is useful for summarizing and analyzing data across multiple dimensions. It's commonly used for financial reporting, sales analysis, and cross-tabulated data analysis.

26. Azure Map Visualization:

- ✓ Allows you to visualize spatial data on a map using Azure Maps service within Power BI.
- ✓ Supports various map types, including bubble maps, filled maps, and shape maps.
- ✓ Enables geospatial analysis and visualization of location-based data.

27. Waterfall Chart:

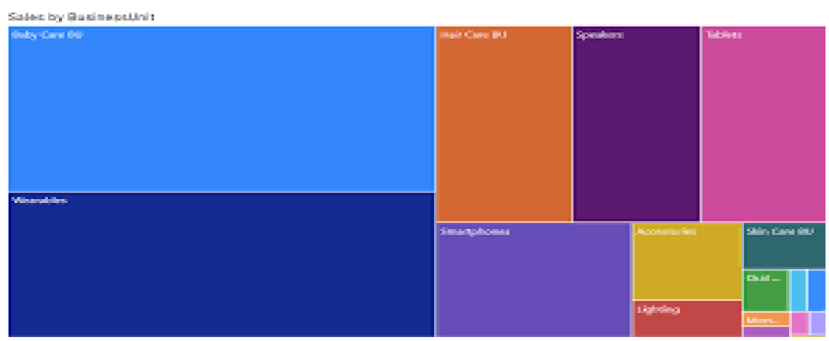
- ✓ Illustrates how an initial value is affected by intermediate positive or negative values, leading to a final value.



- ✓ **Use-**Analyzing changes in profit over a financial period.

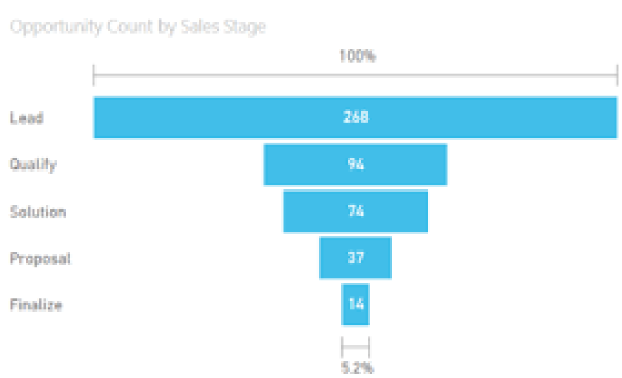
28.Tree Map:

- ✓ Displays hierarchical data as nested rectangles, with the size of each rectangle representing a value.



29. Funnel Chart:

- ✓ Displays stages of a process and the conversion rates between those stages.



- ✓ **Use-**Visuaizing the sales pipeline, from leads to closed deals.

30. R Script Visual:

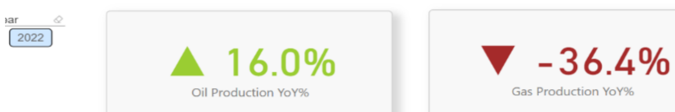
- ✓ Allows you to create custom visualizations using R scripts within Power BI.
- ✓ You can write R code to manipulate and visualize your data, and the results are displayed directly in Power BI reports and dashboards.
- ✓ **Use**-For advanced statistical analysis, custom chart types, and integrating R-based models into Power BI.

31. Python Visual:

- ✓ Similar to the R Script Visual, but allows you to use Python code instead of R.
- ✓ Enables you to leverage Python libraries and tools for data manipulation, analysis, and visualization directly within Power BI.

32. Key Influencer Visual:

- ✓ Analyzes your data to identify key factors or influencers that impact a target variable.
- ✓ Provides insights into the most significant contributors to a particular outcome or metric.



- ✓ Helps users understand the drivers behind trends and patterns in their data.

33. Decomposition Tree Visual:

- ✓ Provides a hierarchical visualization of the decomposition of a measure into its contributing factors.
- ✓ Allows users to explore the breakdown of a metric by different dimensions or attributes.
- ✓ **ssUse**-For analyzing the composition of a metric and understanding the relative importance of its components.

34. Q & A (Question and Answer):

- ✓ Allows users to ask natural language questions about their data and receive instant visualizations and insights.
- ✓ **Use**-Advanced natural language processing (NLP) algorithms to understand and interpret user queries.
- ✓ Provides an intuitive and interactive way for users to explore and analyze data without needing to write complex queries or create visualizations manually.

35. Narrative Visual:

- ✓ Generates dynamic narratives or textual summaries of your data and analysis.
- ✓ Automatically creates written descriptions of key insights, trends, and patterns in your data.
- ✓ Enhances communication and storytelling by providing context and explanations for the data visualizations.

36. Metrics:

- ✓ Represents key performance indicators (KPIs) or metrics in your reports and dashboards.
- ✓ Allows users to track and monitor important business metrics and performance indicators at a glance.
- ✓ Can be visualized using various chart types, such as cards, gauges, or sparklines.

37. Paginated Reports:

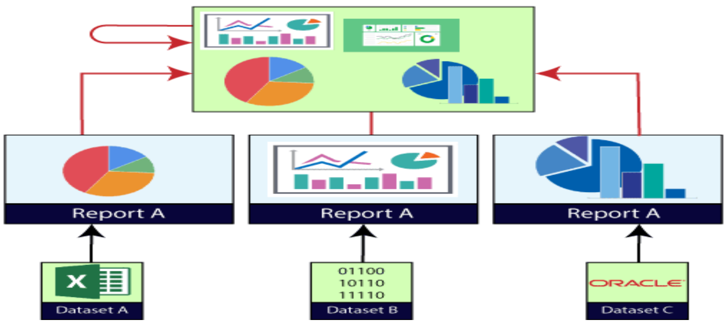
- ✓ Allows you to create pixel-perfect, paginated reports for printing or exporting.
- ✓ Ideal for generating formal reports, invoices, statements, or any other document-style reports.
- ✓ Provides precise control over layout, formatting, and pagination

Power BI Dashboard

Power BI dashboard is a single page, also called a canvas that uses visualization to tell the story. It is limited to one page; therefore, a well-designed dashboard contains only the most essential elements of that story.

The visualizations visible on the dashboard are known as tiles. These tiles are pinned to the dashboard from reports. The visualizations on a dashboard come from reports, and each report is based on one data set.

A dashboard can combine on-premises and cloud-born data. And they are providing a consolidated view regardless of where the data lies.



Creating Dashboard in Power BI

We need to import one sample datasets of the Power BI and use it to create a new dashboard.

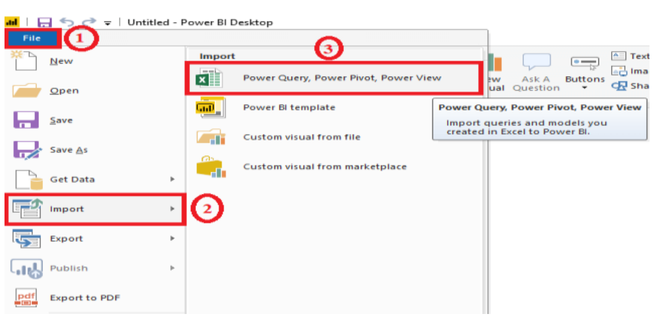
For example, suppose a sample such as Procurement Analysis. This sample is an excel workbook with two PowerView sheets.

When Power BI imports the workbook, it adds a dataset and a report to the workspace. Let's see step by step.

Step 1: Open the Power BI Desktop and click on the File pane.

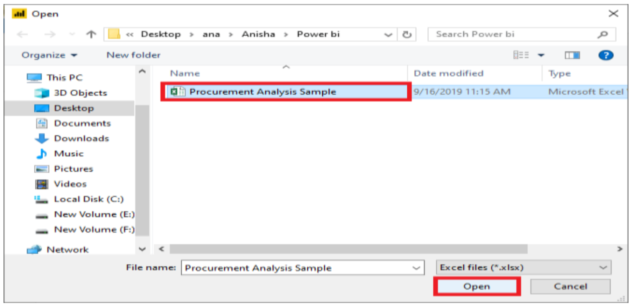
Step 2: Go to the Import option.

Step 3: And select the Excel dataset file to import the file.

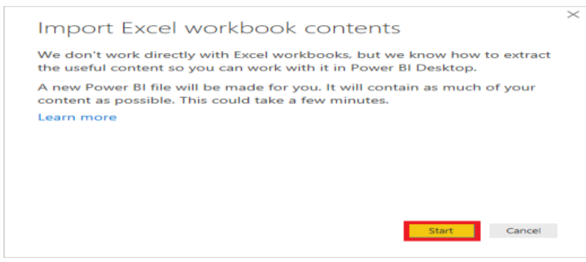


Step 4: Select the procurement analysis sample file.

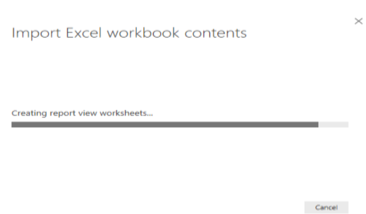
Step 5: And click on the Open button.



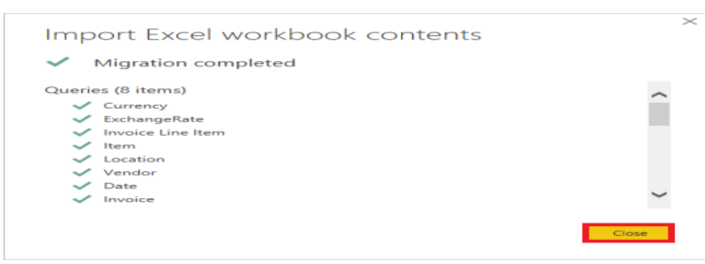
Step 6: For the exercise, select the Start button.



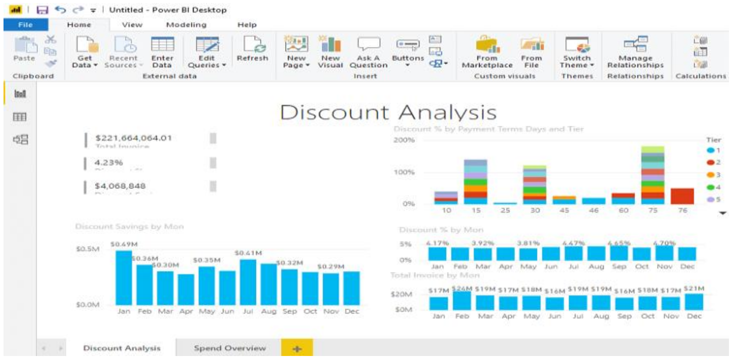
It starts import excel workbook and creating report view worksheets shown in the below screenshot.



Step 7: When the completed message appears, then select the Close button to dismiss it.

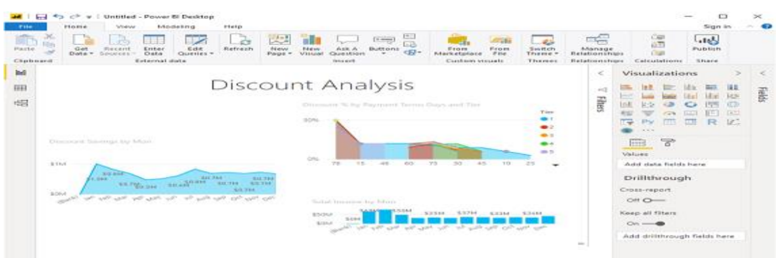


In the below screenshot, you can see the discount analysis of the imported dataset in the form of tiles.



Power BI Reports

A Power BI **report** is a multi-perspective view into the dataset, with visualizations which represent different findings and insights from that dataset.A report can have a single visualization or multiple visualizations. The visualizations in a report represent something like a dashboard does but serve a different purpose.These visualizations are not static. These are highly interactive & highly customizable visualizations which update, as the underlying data changes. You can add and remove the data, change visualization types, and apply filters in your model to discover insights.



Difference between Dashboards and Report

Dashboard and reports both terms are used interchangeably, but they are not synonymous. The below table compares the dashboard with the reports, such as:

Capabilities	Dashboards	Reports
Pages	It has only one page.	It can have one or more pages.

Data Sources	It has one or more reports and datasets per dashboard.	It has only a single dataset per report.
Pinning	It can pin existing visualizations only from the current dashboard to your other dashboards.	It can pin visualizations to any of the dashboards. And also can pin entire report pages to any of the dashboards.
Filtering	It can't filter or slice.	It has many different ways to filter, highlight, and slice.
Feature	It can set one dashboard as the featured dashboard.	It cannot create a feature report.
Set alerts	No	Yes, it can set alerts.
Subscribe	We can't subscribe to a dashboard.	We can subscribe to report pages.
Available in Power BI desktop	No	Yes, it can create and view reports in desktop.
Change visualization type	No, if a report owner changes the visualization type in the report, the pinned visualization on the dashboard does not update.	Yes, it can change the visualization type.

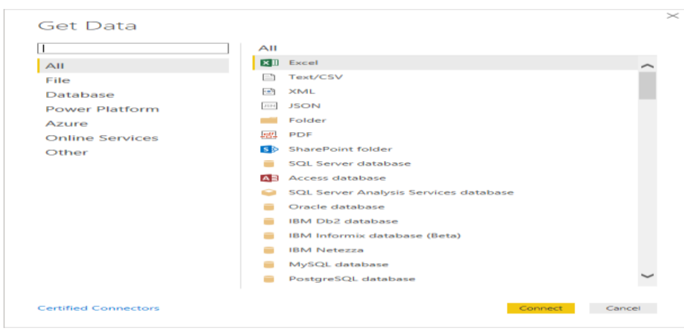
Power BI Data Sources

Power BI Desktop and Power BI Services support a large range of data sources. Click on the **Get Data** button, and it shows you all the available data connections.

You can connect to different **Flat files, Azure cloud, SQL database,** and **Web platforms**, also such as **Google Analytics, Facebook,** and **Salesforce** objects. It includes an ODBC connection to connect to other ODBC data sources.Here are the available data sources in Power BI, as shown below:

- SQL Database
- Flat Files
- Blank Query
- OData Feed
- Azure Cloud Platform
- Online Services
- Oracle database
- IBM Db2 database
- IBM Netezza
- IBM Informix database (Beta)
- Other data sources such as Exchange, Hadoop, or active directory

To connect data in Power BI Desktop, you need to click on the **Get Data** button in the main screen. First, it shows you the most common data sources. Then click on the **More** option to see a full available list of the data sources.



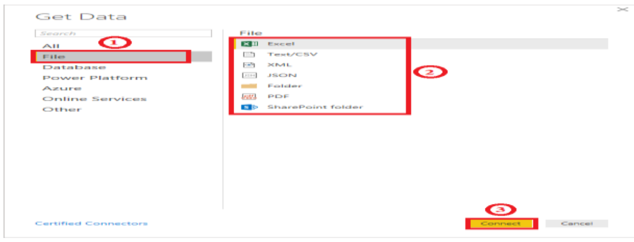
On the left side, it shows a category of all the available data sources. You also have an option to perform search operation at the top.Let's see all the listed data sources in detail:

1. All

In this category, you can see all the available data sources of the Power BI desktop.

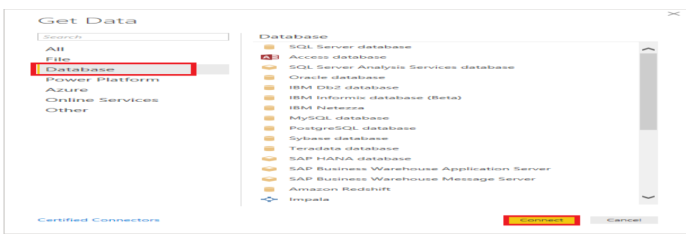
2. File

When you click on the **File** option, it shows you all the flat files supported in Power BI desktop. Select any file type from the list and click on the **Connect** button to connect that file.



3. Database

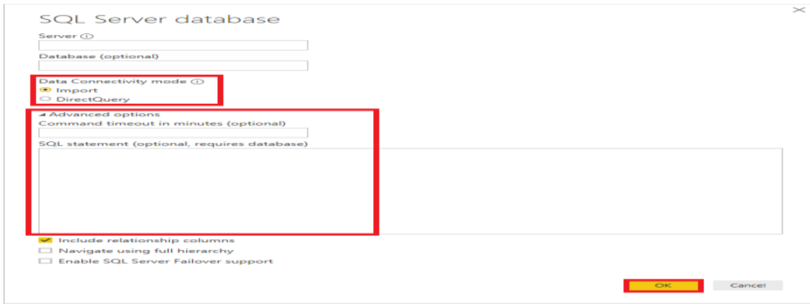
When you click on the **Database** option, it shows you the list of all the database connections that you can connect to any database.



You need to pass the server name, user name, and password to connect. Also, you can connect via a direct SQL query using the **Advanced** option. You can also select connectivity mode **- Import** or **DirectQuery**.

Import: Import method allows to perform data transformations and manipulation. When you publish the data to PBI service (limit 1 GB), it consumes and pushes data into Power BI Azure backend and data can be refreshed up to 8 times a day and a schedule can be set up for data refresh.

DirectQuery: It limits the option of data manipulation, and the data stays in the SQL database. The DirectQuery is live, and there is no need to schedule refresh as in the Import method.



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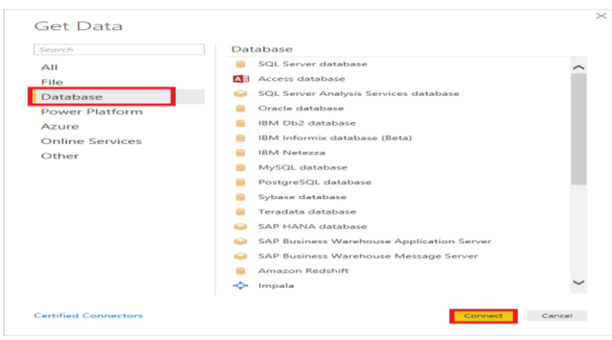

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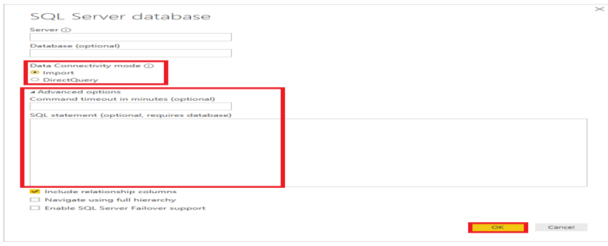
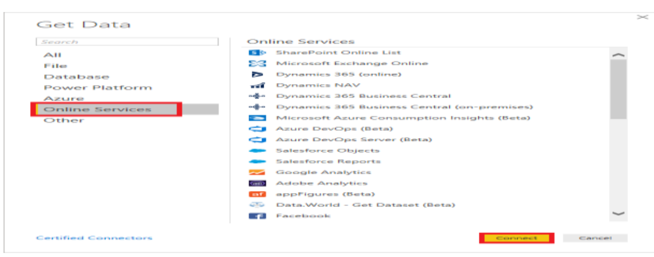


4.Azure

Using the Azure option, you can connect with the database in the Azure cloud. Below screenshot shows you the various options available under the Azure category.

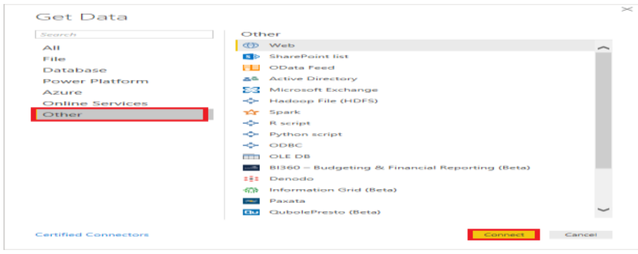
5. Online Services

The Power BI also allows you to connect to different online services such as **Exchange, Salesforce, Google Analytics,** and **Facebook**.Following screenshots showed the various options available under Online Services.



6.Other

Below screenshot shows the various options available under other categories.



DASHBOARD

A dashboard in Power BI is a single-page canvas that contains multiple visuals (charts, graphs, tables, etc.) representing key metrics, trends, and insights from your data. It provides a consolidated view of information, allowing users to monitor performance, analyze data, and gain insights at a glance. Here's how you can create a dashboard in Power BI:

1. Build Your Visualizations:

- ✓ Create or import the necessary reports and datasets into Power BI Desktop.
- ✓ Design and customize the visuals (charts, graphs, tables, etc.) that you want to include in your dashboard.
- ✓ Ensure that your visuals are well-designed, interactive, and provide valuable insights into your data.

2. Arrange Visuals on the Canvas:

- ✓ Open a new or existing dashboard in Power BI Desktop.
- ✓ Drag and drop the visuals from your reports onto the dashboard canvas.
- ✓ Arrange and resize the visuals to create an organized and visually appealing layout.

3. Add Interactivity:

- ✓ Configure interactivity features such as slicers, filters, and drill-down functionality to allow users to explore and interact with the data dynamically.
- ✓ Add buttons or bookmarks to enable navigation between different sections or views within the dashboard.

4. Customize Layout and Formatting:

- ✓ Customize the layout, formatting, and styling of your dashboard to align with your organization's branding and design guidelines.
- ✓ Use themes, color palettes, and fonts to create a cohesive and visually appealing aesthetic.

5. Publish and Share:

- ✓ Save your dashboard in Power BI Desktop and publish it to the Power BI service.
- ✓ Share the dashboard with intended users or groups within your organization by granting access permissions.
- ✓ Optionally, embed the dashboard into websites, applications, or other platforms to make it accessible to a wider audience.

6. Monitor and Analyze:

- ✓ Monitor usage metrics and performance indicators to track engagement and identify areas for improvement.
- ✓ Analyze user feedback and iterate on your dashboard design and content based on user needs and preferences.

DASHBOARD CREATING

Creating a dashboard in Power BI involves consolidating key visuals and insights from your reports onto a single canvas for easy monitoring and analysis. Here's a step-by-step guide to creating a dashboard in Power BI:

1. Prepare Your Visuals:

- ✓ Design and build the reports containing the visuals you want to include in your dashboard.
- ✓ Ensure that your visuals are well-designed, interactive, and provide valuable insights into your data.

2. Open Power BI Desktop:

- ✓ Launch Power BI Desktop and open the report(s) containing the visuals you want to add to your dashboard.

3. Add Visuals to the Dashboard:

- ✓ Navigate to the report page containing the visual you want to add to your dashboard.

- ✓ Click on the visual to select it, then click on the "Pin Visual" icon in the top-right corner of the visual.
- ✓ Choose the option to pin the visual to a new or existing dashboard.
- ✓ Repeat this process for each visual you want to add to your dashboard.

4. Arrange and Customize:

- ✓ Open the dashboard pane by clicking on the "Dashboard" icon in the left navigation pane.
- ✓ Drag and drop the pinned visuals onto the dashboard canvas to arrange them as desired.
- ✓ Resize and customize the visuals as needed to create an organized and visually appealing layout.
- ✓ Add additional elements such as text boxes, shapes, and images to provide context and enhance the dashboard's visual appeal.

5. Add Interactivity:

- ✓ Configure interactivity features such as slicers, filters, and drill-down functionality to allow users to explore and analyze the data dynamically.
- ✓ Use bookmarks or buttons to enable navigation between different sections or views within the dashboard.

6. Customize Formatting:

- ✓ Customize the layout, formatting, and styling of your dashboard to align with your organization's branding and design guidelines.
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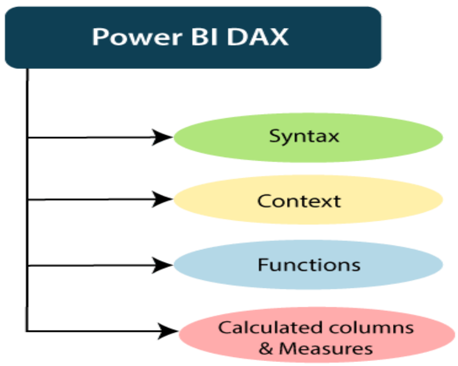
POWER BI DAX

DAX (Data Analysis Expressions) is a formula language used in Power BI, Excel Power Pivot, and SQL Server Analysis Services (SSAS) Tabular models for data modeling and calculations. It enables you to create calculated columns, measures, and queries to perform data analysis and manipulation. Here's an overview of DAX and its key components:



How does it work?

For understanding the Power BI DAX, it has main three fundamental concepts such as:

- Syntax
- Context
- Functions

1. Calculated Columns:

- ✓ Description: Calculated columns are columns computed based on expressions defined in DAX. They are stored in the data model and can be used like any other column in your dataset.

Syntax:

makefile

Copy code

```
NewColumn = <expression>
```

Example:

css

Copy code

```
Profit = Sales[Revenue] - Sales[Cost]
```

2. Measures:

- ✓ Description: Measures are calculations performed on your data dynamically at query time. They are typically used for aggregations, such as sums, averages, counts, etc.

Syntax:

lua

Copy code

```
MeasureName = <aggregate function>(<column>)
```

Example:

scss

Copy code

```
Total Sales = SUM(Sales[Revenue])
```

3. Filter Context and Row Context:

- ✓ Filter Context: It is the set of filters applied to the data model, such as slicers, filters, and cross-filtering. DAX expressions evaluate within the context of these filters.
- ✓ Row Context: It is the current row being evaluated in a table. DAX expressions automatically iterate over each row in a table to compute results.

4. DAX Functions:

- ✓ Aggregate Functions: SUM, AVERAGE, COUNT, MIN, MAX, etc.
- ✓ Logical Functions: IF, SWITCH, AND, OR, NOT, etc.

- ✓ Statistical Functions: AVERAGEX, SUMX, MINX, MAXX, etc.
- ✓ Date and Time Functions: DATE, YEAR, MONTH, DAY, CALENDAR, etc.
- ✓ Text Functions: CONCATENATE, LEFT, RIGHT, MID, etc.
- ✓ Relationship Functions: RELATED, RELATEDTABLE, CROSSFILTER, etc.

5. Iterators:

- ✓ Description: Iterators apply a function or expression to each row in a table or a specific set of rows.
- ✓ Examples:
- ✓ SUMX: Summarizes the results of an expression evaluated for each row in a table.
- ✓ FILTER: Returns a table that contains only the rows that meet specified criteria.

6. Context Transition:

- ✓ Description: Context transition occurs when a DAX expression transitions from row context to filter context, or vice versa. Understanding context transition is crucial for writing accurate and efficient DAX expressions.

7. Time Intelligence Functions:

- ✓ Description: Time intelligence functions facilitate analysis over time-based data, such as year-to-date calculations, moving averages, and year-over-year comparisons.

Examples:

TOTALYTD: Calculates the year-to-date total for a given expression.

SAMEPERIODLASTYEAR: Returns a table of dates for the same period in the previous year.

8. DAX Date Functions:

- ✓ DATEADD: Adds or subtracts a specified number of units (days, months, quarters, years) to a given date.
- ✓ Example: DATEADD(Calendar[Date], -1, MONTH)
- ✓ DATESBETWEEN: Returns a table of dates that fall between two specified dates.

- ✓ Example: DATESBETWEEN(Calendar[Date], DATE(2022,1,1), DATE(2022,12,31))
- ✓ TOTALMTD: Calculates the total of an expression from the beginning of the month to the current date.
- ✓ Example: TOTALMTD(SUM(Sales[Revenue]), Calendar[Date])

9. Aggregators in DAX:

- ✓ SUM: Calculates the sum of values in a column or expression.
- ✓ Example: Total Sales = SUM(Sales[Amount])
- ✓ AVERAGE: Calculates the average of values in a column or expression.
- ✓ Example: Average Price = AVERAGE(Products[Price])
- ✓ COUNT: Counts the number of rows in a table or the number of non-blank values in a column.
- ✓ Example: Total Orders = COUNT(Orders[OrderID])

10. Rolling Totals and Cumulative Sums:

- ✓ Running Total: Calculates a cumulative total over a specified period.
- ✓ Example: Running Total = CALCULATE(SUM(Sales[Revenue]), FILTER(ALL(Calendar), Calendar[Date] <= MAX(Calendar[Date])))
- ✓ Moving Average: Calculates an average over a moving window of a specified size.
- ✓ Example: Moving Average = AVERAGEX(FILTER(Calendar, Calendar[Date] <= EARLIER(Calendar[Date]) && Calendar[Date] >= EARLIER(Calendar[Date]) - 30), [SalesAmount])

11. Financial Calculations:

- ✓ NPV (Net Present Value): Calculates the net present value of an investment based on a series of cash flows.
- ✓ Example: NPV = NPV(0.1, [Cash Flow 1], [Cash Flow 2], ...)
- ✓ IRR (Internal Rate of Return): Calculates the internal rate of return of an investment based on a series of cash flows.
- ✓ Example: IRR = IRR([Cash Flow 1], [Cash Flow 2], ...)

12. Today's Date Calculation:

- ✓ To calculate today's date in DAX, you can use the TODAY() function. This function returns the current date based on the system date and time settings of the computer running Power BI.

dax

Copy code

Today_Date = TODAY()

2. Yesterday's Date Calculation:

- ✓ To calculate yesterday's date in DAX, you can subtract one day from today's date using the TODAY() function in combination with the DATEADD() function.

  dax

  Copy code

  Yesterday_Date = TODAY() - 1

  Alternatively, you can use the EDATE() function to subtract one day from today's date:

  dax

  Copy code

  Yesterday_Date = EDATE(TODAY(), -1)

- ✓ Example:

  Let's say you have a table named DateTable with a column named Date. You can create calculated columns for today's and yesterday's date as follows:

  dax

  Copy code

  DateTable[Today_Date] = TODAY() DateTable[Yesterday_Date] = TODAY() - 1

  Note:

  Remember that the TODAY() function returns the current date based on the system date and time settings of the computer running Power BI. If your report is being viewed in a different time zone, the results may differ.

It's important to consider timezone differences and any requirements specific to your data and reporting scenario when working with date calculations in Power BI.

PREPROCESSING

Preprocessing in Power BI involves cleaning, transforming, and preparing data for analysis and visualization. Here are some common preprocessing steps you can perform in Power BI:

1. Data Loading:

- ✓ Connect to Data Sources: Import data from various sources such as Excel files, databases, online services, or cloud storage.
- ✓ Data Refresh: Schedule automatic data refreshes to keep your dataset up-to-date.

2. Data Cleaning:

- ✓ Handle Missing Values: Remove or replace missing values in your dataset to ensure data integrity.
- ✓ Remove Duplicates: Eliminate duplicate records from your dataset to avoid redundancy.
- ✓ Standardize Data: Convert data into a consistent format, such as converting dates to a standard date format.

3. Data Transformation:

- ✓ Data Splitting: Split columns containing combined data into separate columns.
- ✓ Data Merging: Merge multiple datasets based on common fields.
- ✓ Data Filtering: Filter out irrelevant data or rows that do not meet specific criteria.
- ✓ Data Aggregation: Aggregate data by grouping rows and calculating summary statistics.

4. Data Enrichment:

- ✓ Add Calculated Columns: Create new columns by performing calculations on existing data, such as adding a calculated profit column.
- ✓ Data Lookup: Enhance your dataset by adding additional information from lookup tables or external sources.

5. Data Modeling:

- ✓ Define Relationships: Establish relationships between tables in your dataset to enable cross-table analysis.
- ✓ Create Hierarchies: Define hierarchical structures to facilitate drill-down analysis.

6. Data Formatting:

- ✓ Format Data Types: Ensure that data types are correctly assigned to each column (e.g., text, number, date).
- ✓ Format Values: Format values to improve readability, such as applying currency symbols or date formats.

7. Data Sampling and Splitting:

- ✓ Data Sampling: Analyze a subset of your data to speed up performance or focus on specific segments.
- ✓ Data Splitting: Split your dataset into training and testing sets for machine learning or predictive analytics.

8. Data Security:

- ✓ Apply Row-Level Security: Restrict access to certain rows of data based on user roles or permissions.
- ✓ Data Encryption: Encrypt sensitive data to protect it from unauthorized access.

9. Data Profiling and Exploration:

- ✓ Data Profiling: Analyze your data to understand its structure, quality, and distribution.
- ✓ Data Exploration: Explore your data visually using charts and graphs to uncover insights and trends.

Navigator

In Power BI, the "Navigator" refers to the interface used to connect to and import data from external data sources. When you open Power BI Desktop and choose to get data, you'll be presented with the Navigator window. Here's an overview of how the Navigator works:

1. Accessing the Navigator:

- ✓ To access the Navigator, click on the "Home" tab in Power BI Desktop.
- ✓ Then, click on the "Get Data" button in the "External Data" group.

- ✓ This will open the Navigator window where you can choose the data source you want to connect to.

2. Selecting a Data Source:

- ✓ In the Navigator window, you'll see a list of available data sources categorized by type (e.g., databases, files, online services).
- ✓ Click on the data source you want to connect to. For example, you might choose "Excel" if your data is stored in an Excel file, or "SQL Server" if your data is in a SQL Server database.

3. Connecting to the Data Source:

- ✓ After selecting a data source, you may need to provide additional information such as the file path, server name, database name, or credentials.
- ✓ Enter the required information and click "Connect" to establish a connection to the data source.

4. Previewing and Selecting Data:

- ✓ Once connected, you'll see a preview of the data available in the data source.
- ✓ You can browse through the tables or files available and select the specific data you want to import into Power BI.
- ✓ Click on a table or file to select it, then click "Load" or "Transform Data" to import the data into Power BI.

5. Data Import Options:

- ✓ When importing data, you have the option to load it directly into Power BI ("Load") or perform additional transformations using Power Query Editor ("Transform Data").
- ✓ Power Query Editor allows you to clean, transform, and reshape your data before loading it into Power BI.

6. Refreshing Data:

- ✓ After importing data into Power BI, you can configure automatic data refresh settings to keep your dataset up-to-date.
- ✓ Power BI can automatically refresh data at regular intervals or on-demand, depending on your preferences.

Bookmark

In Power BI, bookmarks allow you to capture the current state of your report, including the selection, visibility, and formatting of visuals, and save it for future reference or interaction. Here's an overview of how bookmarks work in Power BI:

1.Creating Bookmarks:

- ✓ Capturing State: Arrange the visuals on your report canvas and apply any filters, slicers, or interactions as desired.
- ✓ Creating Bookmark: Go to the "View" tab in Power BI Desktop and click on "Bookmarks" to open the Bookmarks pane. Then, click on "Add" to create a new bookmark.
- ✓ Name and Description: Give your bookmark a meaningful name and description to help you identify it later.
- ✓ Options: Choose the options you want to include in the bookmark, such as the selection state, visibility state, and display mode of visuals.
- ✓ Apply: Click on "Apply" to save the bookmark.

2. Managing Bookmarks:

- ✓ Rename: Double-click on a bookmark in the Bookmarks pane to rename it.
- ✓ Delete: Select a bookmark and click on the "Delete" icon to remove it.
- ✓ Update: After making changes to your report, you can update an existing bookmark by selecting it and clicking on the "Update" icon.

3. Using Bookmarks:

- ✓ Selection Pane: Bookmarks can be applied from the Bookmarks pane by clicking on them.
- ✓ Buttons: You can also create buttons on your report canvas and assign bookmarks to them as actions. This allows users to interactively navigate between different bookmarked states.
- ✓ Visual Interactions: Bookmarks can be used to control the visibility and interaction of visuals. For example, you can create a bookmark to show or hide specific visuals based on user preferences.

4. Best Practices:

- ✓ Organize: Use meaningful names and descriptions for bookmarks to make them easy to understand and manage.
- ✓ Test: Test your bookmarks to ensure they capture the desired state of your report accurately.
- ✓ Iterate: Iterate on your bookmarks as you refine and enhance your report design and user experience.

GROUPING

In Power BI, grouping in reports refers to organizing visuals and other report elements into logical groups for better organization and navigation. Here's how you can use grouping in Power BI reports:

1. Grouping Visuals:

- ✓ Select Visuals: Click and drag to select multiple visuals on the report canvas.
- ✓ Group Visuals: Right-click on one of the selected visuals and choose "Group" from the context menu.
- ✓ Name Group: Give your group a meaningful name to describe the content or purpose of the grouped visuals.
- ✓ Adjust Group: You can resize, move, or format the group container as needed.

2. Ungrouping Visuals:

- ✓ To ungroup visuals, right-click on the group container and choose "Ungroup."
- ✓ Using Groups for Navigation:
- ✓ Groups can act as containers for related visuals, making it easier to organize and navigate through complex reports.
- ✓ You can collapse or expand groups to show or hide the visuals they contain, allowing users to focus on specific sections of the report.

3. Best Practices:

- ✓ Logical Grouping: Group visuals that are related or represent similar data to improve the clarity and organization of your report.
- ✓ Consistency: Maintain consistent naming conventions and formatting for groups to enhance readability and usability.
- ✓ Hierarchy: Use nested groups to create hierarchical structures for more complex reports with multiple levels of organization.

- ✓ Testing: Test your report layout and grouping to ensure it provides an intuitive and user-friendly experience for consumers.

REPORTS

In Power BI, reports serve as the primary means of presenting data visualizations, analysis, and insights to users. Here's an overview of creating and working with reports in Power BI:

1. Creating a Report:

- ✓ Data Import: Connect to one or more data sources and import the necessary data into Power BI Desktop.
- ✓ Visualizations: Drag and drop fields from your dataset onto the report canvas to create visualizations such as charts, tables, and maps.
- ✓ Interactivity: Add slicers, filters, and other interactive elements to allow users to explore and analyze the data dynamically.

2. Designing and Formatting:

- ✓ Layout: Arrange visualizations on the canvas to create an intuitive and appealing layout.
- ✓ Formatting: Customize visualizations by adjusting colors, fonts, labels, and other formatting options to enhance readability and visual appeal.
- ✓ Themes: Apply pre-defined or custom themes to maintain consistency and branding across reports.

3. Adding Interactivity:

- ✓ Slicers: Add slicers to allow users to filter data dynamically by selecting specific values or ranges.
- ✓ Drill-down: Enable drill-down functionality to allow users to explore data at different levels of detail.
- ✓ Bookmarks: Create bookmarks to capture and save specific views or states of the report for easy reference or navigation.
- ✓ Buttons: Add buttons with actions such as bookmark navigation, report page navigation, or URL links to enhance interactivity.

4. Collaboration and Sharing:

- ✓ Publishing: Publish your report to the Power BI service to share it with others in your organization or with external stakeholders.
- ✓ Sharing: Share reports with specific users or groups within your organization and control their access permissions.
- ✓ Embedding: Embed reports into websites, applications, or other platforms to make them accessible to a wider audience.

5. Monitoring and Governance:

- ✓ Usage Metrics: Monitor report usage and performance metrics to track engagement and identify areas for improvement.
- ✓ Data Refresh: Configure automatic data refresh schedules to keep your reports up-to-date with the latest data.
- ✓ Data Security: Implement row-level security and other access controls to restrict access to sensitive data within reports.

6. Iteration and Improvement:

- ✓ Feedback: Gather feedback from users and stakeholders to identify areas for improvement and iterate on report design and content.
- ✓ Data Modeling: Continuously refine data models and calculations to ensure accuracy and relevance of insights.
- ✓ Versioning: Maintain version history and track changes to reports over time to facilitate collaboration and traceability.

MODELING

Modeling in Power BI involves shaping and transforming your data to create a logical structure that facilitates analysis and visualization. Here's an overview of key aspects of modeling in Power BI:

1. Data Modeling:

- ✓ Data Import: Connect to data sources and import data into Power BI.
- ✓ Table Relationships: Define relationships between tables based on common fields to establish connections and enable cross-table analysis.
- ✓ Data Types: Define appropriate data types for each column in your dataset to ensure accurate calculations and formatting.

- ✓ Calculated Columns: Create new columns by defining calculated expressions based on existing data in your dataset.

2. DAX (Data Analysis Expressions):

- ✓ DAX Functions: Use DAX functions to perform calculations, aggregation, filtering, and manipulation of data.
- ✓ Measures: Define measures to calculate aggregate values such as sums, averages, counts, and ratios.
- ✓ Calculated Tables: Create calculated tables based on DAX expressions to generate new tables dynamically within your data model.

3. Hierarchies and Groups:

- ✓ Hierarchical Structures: Define hierarchical relationships between fields to enable drill-down analysis and navigation.
- ✓ Grouping: Group related fields or values together to simplify analysis and visualization.

4. Data Transformations:

- ✓ Power Query Editor: Use Power Query Editor to perform data cleansing, transformation, and shaping operations.
- ✓ Merging and Appending: Merge or append data from multiple tables to create consolidated datasets for analysis.
- ✓ Data Splitting: Split columns containing combined or nested data into separate columns for better analysis.

5. Advanced Modeling Techniques:

- ✓ Role-Playing Dimensions: Handle scenarios where a single dimension table is used in multiple roles within a data model.
- ✓ Time Intelligence: Use DAX functions for time intelligence to perform analysis over time periods, such as year-to-date calculations, moving averages, and year-over-year comparisons.
- ✓ Advanced Relationships: Implement advanced relationship types such as many-to-many relationships and bidirectional filtering to model complex data scenarios.

6. Optimization and Performance:

- ✓ Data Model Optimization: Optimize your data model by reducing unnecessary columns, tables, or relationships to improve performance.
- ✓ Query Optimization: Optimize DAX expressions and queries to minimize computational overhead and improve responsiveness.
- ✓ Data Compression: Utilize data compression techniques to reduce the size of your data model and improve performance.

7. Iteration and Refinement:

- ✓ Testing: Test your data model and calculations to ensure accuracy and reliability.
- ✓ Iterative Development: Iterate on your data model based on user feedback and changing business requirements.
- ✓ Documentation: Document your data model, calculations, and assumptions to facilitate collaboration and knowledge sharing.

APP CREATION

Creating an app in Power BI involves packaging and organizing your reports, dashboards, and datasets into a unified experience for end-users. Here's a step-by-step guide to creating an app in Power BI:

1. Prepare Your Content:

- ✓ Reports: Design and build the reports and dashboards you want to include in your app. Ensure they are well-designed, interactive, and provide valuable insights.
- ✓ Datasets: Import and prepare the datasets needed for your reports and dashboards. Perform any necessary data modeling and transformations to ensure data accuracy and relevance.

2. Create Workspaces:

- ✓ Organize Content: Group related reports, dashboards, and datasets into separate workspaces based on their intended audience or purpose.
- ✓ Access Control: Set access permissions for each workspace to control who can view, edit, or publish content within the workspace.

3. Package Content into an App:

- ✓ Navigate to Apps: In the Power BI service, go to the "Apps" section in the navigation pane.

- ✓ Create New App: Click on "Create" or "New app" to start creating a new app.
- ✓ Select Content: Choose the reports, dashboards, and datasets you want to include in your app. You can select content from multiple workspaces.
- ✓ Customize Settings: Configure app settings such as display name, description, cover image, and navigation options.

4. Configure Permissions:

- ✓ Access Permissions: Specify who can access the app by defining access permissions. You can grant access to specific users, groups, or distribution lists.
- ✓ Install or Publish: Choose whether to install the app directly for specific users or publish it to the Power BI service for broader availability.

5. Publish and Share the App:

- ✓ Publish App: Once configured, publish the app to make it available to users. Users with appropriate permissions will be able to install and access the app from the Apps section in Power BI.
- ✓ Share App Link: Share the app link with intended users or groups to notify them of its availability.

6. Monitor Usage and Feedback:

- ✓ Usage Metrics: Monitor app usage and performance metrics to track engagement and identify areas for improvement.
- ✓ Feedback Collection: Gather feedback from users to understand their needs and preferences, and iterate on app design and content accordingly.

7. Iterate and Improve:

- ✓ Iterative Development: Continuously update and improve your app based on user feedback, changing business requirements, and emerging trends.
- ✓ Version Control: Maintain version history and track changes to your app over time to facilitate collaboration and traceability.

PUBLISHING

Publishing in Power BI involves sharing your reports, dashboards, and datasets with others within your organization or with external stakeholders. Here's a step-by-step guide to publishing your content in Power BI:

1. Prepare Your Content:

   - ✓ Reports and Dashboards: Design and build the reports and dashboards you want to share. Ensure they are well-designed, interactive, and provide valuable insights.
   - ✓ Datasets: Import and prepare the datasets needed for your reports and dashboards. Perform any necessary data modeling and transformations to ensure data accuracy and relevance.

2. Save Your Work:

   - ✓ Save to Power BI Service: In Power BI Desktop, click on the "File" menu and select "Save As." Choose "Power BI Service" as the destination to save your report to the Power BI service.

3. Publish to Power BI Service:

   - ✓ Navigate to Power BI Service: Open a web browser and go to the Power BI service (https://app.powerbi.com).
   - ✓ Sign In: Sign in to your Power BI account using your credentials.
   - ✓ Navigate to Workspaces: Go to the "Workspaces" section in the left navigation pane.
   - ✓ Select Workspace: Choose the workspace where you want to publish your content or create a new workspace if needed.
   - ✓ Import Content: Click on the "Import" button and select "File" to upload your Power BI Desktop file (.pbix) containing your report, dashboard, and dataset.
   - ✓ Configure Settings: Configure settings such as dataset refresh schedule, permissions, and options for integrating with other Office 365 services if necessary.
   - ✓ Publish: Click on the "Publish" button to upload your content to the Power BI service.

4. Share Your Content:

   - ✓ Access Permissions: Specify who can access your content by setting access permissions for the workspace. You can grant access to specific users, groups, or distribution lists.
   - ✓ Share Links: Share the links to your reports, dashboards, or workspaces with intended users or groups to provide access.

5. Collaborate and Iterate:

- ✓ Collaborate: Collaborate with other users or stakeholders by allowing them to view, edit, or comment on your reports and dashboards.
- ✓ Iterate: Continuously update and improve your content based on feedback, changing business requirements, and emerging trends.

6. Monitor Usage and Performance:

- ✓ Usage Metrics: Monitor usage metrics such as views, interactions, and refresh history to track engagement and identify areas for improvement.
- ✓ Performance Monitoring: Monitor performance metrics to ensure that your reports and datasets are responsive and reliable.

7. Governance and Security:

- ✓ Data Security: Implement row-level security and other access controls to restrict access to sensitive data within reports and datasets.
- ✓ Compliance: Ensure compliance with data privacy regulations and organizational policies regarding data handling and sharing.